

Accelerating Recovery

What sets great climbers apart from others is not their physical prowess (amazing as it may be) but their brains. Peak performers (of all kinds) think and act in unique ways.

If you are serious about climbing performance, then you must be serious about accelerating recovery. Knowing how to limit fatigue and speed recovery is as important as knowing how to perform a drop-knee, lock a finger jam, or float a deadpoint. The bottom line: If you are not playing a proactive role in the recovery process, then you are definitely not training optimally or climbing up to your capability.

We all know, firsthand, that physical fatigue is a primary limiting factor whether pulling down at the crags or training in the gym. Therefore, it stands to reason that being able to accelerate recovery means you will get more back during a mid-climb shakeout and while resting between climbs and days of climbing. As a result, you will perform better on the rocks today, tomorrow, and on all your future outings. Similarly, more rapid recovery between training sessions can translate into more long-term strength gains, since you can work out more often while still getting adequate rest and without risk of overtraining.

While many of today's enthusiastic climbers are keen on staying current on the latest climbing and training techniques, surprisingly few individuals are aware of the numerous strategies for accelerating recovery. Recovery from exercise has been the subject of dozens of recent research studies, and any serious climber would be wise to

heed the findings of these sports scientists. On the pages that follow, I will present the leading-edge recovery strategies used in these studies, as well as provide instruction on specific techniques that will help you slow fatigue and speed recovery while on the rock.

Clearly, recovery ability is a function of several factors, including age, sex, and level of conditioning. Regardless of these factors, however, I guarantee that you can recover more quickly by playing a proactive role in the recovery process—instead of just letting it happen, as many climbers do. By placing the same importance on optimal recovery as you do on training optimally, you will enhance your training response as well as your overall climbing performance!

The Basics of Fatigue and Recovery

Proactively managing fatigue and taking the steps to accelerate recovery require an understanding of the basic physiological processes involved in energy production, fatigue, and recovery. Chapter 5 presented a detailed study of energy systems and fatigue, but let's recap the key points before getting into specific recovery techniques.

Causes of Fatigue

Several factors contribute to the fatigue you experience while training or climbing. These factors include the depletion of muscle fuels, the accumulation of metabolic by-products, low blood glucose, muscular cramps and microtraumas, and, finally, central fatigue.

DEPLETION OF ATP-CP

Adenosine triphosphate (ATP) and creatine phosphate (CP) are energy-rich phosphate compounds stored within the muscle cells in small amounts. Brief, maximum-intensity activities (such as a short, vicious boulder problem, a one-arm pull-up, or a 100-meter sprint) are fueled by ATP and CP; the supply of these fuels, however, limits this action to between five and twelve seconds.

This constraint on power output explains why it's next to impossible to perform double dyno campus training for more than about ten to twelve seconds, or why you have less than about ten seconds to pull a maximal move (for you) on a route before your muscles give out. Continued exercise beyond this time threshold is only possible by lowering the intensity of the activity so that the glycolytic energy system can handle the energy production.

Fortunately, ATP and CP are continually synthesized within the muscles, and ATP-CP stores become fully replenished in just three to five minutes of complete rest (Bloomfield 1994).

ACCUMULATION OF METABOLIC BY-PRODUCTS

Sustained high-intensity activity lasting between twelve seconds and two minutes is fueled primarily by the anaerobic metabolism of glycogen fuel (see table 5.3). Unfortunately, metabolic by-products of this energy production—chiefly, hydrogen ions (H^+)—result in muscular discomfort and, eventually, muscular failure. How long you can exercise during periods of rising intracellular acidosis (and hypoxia) depends on your intra- and extracellular buffering capacity and the broader capability of your body to metabolize the numerous by-products of anaerobic glycolysis.

Strength/power-endurance training will increase your tolerance to acidosis as well as improve cellular and systemic buffering of H^+ . Furthermore, anything you do to enhance blood

flow through the working muscles will help disperse excess lactate to lesser-stressed muscles, where it's consumed for aerobic energy production, and your liver, where it's converted back to glucose. Sustained high-intensity exercise (with no rest), however, will cause acidosis and lactate levels to skyrocket above tolerable levels, and, hence, muscular failure will occur in one to two minutes (sooner if you're truly climbing near your limit). This explains why the “pump clock” runs out in less than two minutes on sustained, near-maximum (for you) crux sequences.

During rest periods, the clearance time of H^+ , lactate, and other metabolic by-products can be anywhere from ten to sixty minutes, depending on the initial level of the H^+ and lactate accumulation and whether the rest is active or passive (more on this later).

DEPLETION OF GLYCOGEN

Steady, long-term exercise typically depletes glycogen stores in ninety minutes to two hours. Running out of glycogen causes the infamous hitting-the-wall phenomenon in marathon running, and it is a contributing factor to your inability to climb hard toward the end of a long day on the rock.

Fortunately, climbing is a stop-and-go activity, so a full two-hour supply of glycogen can be stretched out to last nearly all day. You can also spare your glycogen supplies through regular consumption of additional fuel (sports drinks and foodstuffs) throughout the day. Research implies that carbohydrate feeding during exercise can help extend your glycogen supply by 25 to 50 percent (Coyle 1984).

Your starting level of glycogen is also a crucial factor in determining how long and hard you will be able to climb. If you are climbing for a second or third straight day, you will likely have less than the full ninety-minute to two-hour supply of glycogen. This is because complete replenishment of glycogen stores takes twenty-four hours—a good dinner and a full night's sleep are not enough to restock the

supply completely. When climbing on successive days, it's therefore vital to consume more calories throughout the day (to spare glycogen)—and even so, you may hit the wall a bit sooner on day two than you did on day one.

LOW BLOOD GLUCOSE

Blood glucose (sugar) is but one of the possible fuel sources for working muscles, but it's the only fuel source available to the brain and nervous system. As glycogen supplies dwindle during long-duration activity, the working muscles become increasingly reliant on blood glucose for fuel. As a result, blood glucose levels drop and increasing levels of exhaustion and mental fatigue set in.

As mentioned above, ingestion of carbohydrates will help delay this fatigue by helping to maintain an adequate level of blood glucose.

MUSCLE CRAMPS AND MICROTRAUMAS

Muscle cramps and microtraumas can contribute to the sense of muscular fatigue, though in somewhat different time frames. Muscle cramps typically occur near the end of an exhausting period of muscle action—for instance, when some of the muscles in your back or arms lock up after a long, strenuous session of jamming or upon reaching complete exhaustion in the midst of a long sport climb. In such an instance, twenty to thirty minutes of rest, gentle stretching, and massage, as well as consuming some fluids, will help alleviate the cramping and restore normal muscular function.

Microtraumas are a primary cause of the all-too-common delayed-onset muscle soreness (DOMS). This muscle soreness, which becomes evident from twenty-four to forty-eight hours after strenuous exercise, is a result of microscopic muscle tissue tears and the accompanying tightness and swelling (and, no, this day-after muscle soreness is not caused by “lactic acid,” per the common myth). Strength will be diminished for as long as pain persists, possibly as long as two to five days.

CENTRAL FATIGUE

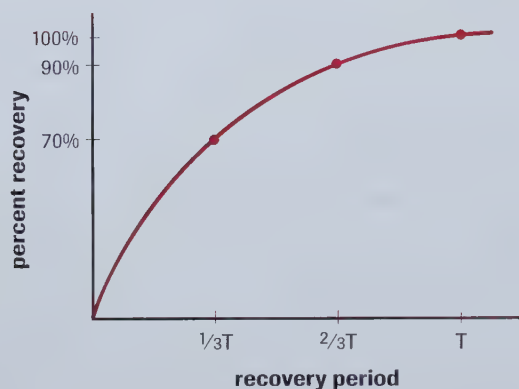
Unlike muscular (peripheral) fatigue, central fatigue is brain-based—think of this as the brain's self-protective central governor limiting the amount of stress (and damage) you can levy on yourself. This so-called central fatigue is realized through a reduction in neural drive (recruitment), impaired coordination, decreased concentration and focus, and diminishing ability to execute difficult motor skills or maximal movement. Repeated high-intensity movements such as lunging and campus training are the hardest on the central nervous system. However, excessive amounts of any specific training stimulus—fingerboard, Hypergravity Isolation Training, campus training, and such, or performing the same type of bouldering movement over and over, can also produce central fatigue. Central fatigue is often most noticeable after back-to-back-to-back days of hard bouldering or projecting; a generalized sense of malaise and the inability to summon peak strength and power, even after days of rest, is the hallmark of central fatigue.

Unfortunately, severe central fatigue can take longer to recover from than that due to any of the other causes. Consider that recovery of a nerve cell takes up to seven times longer than a muscle cell (Bompa 1983). Of course, this level of fatigue may never be experienced by the typical recreational climber, whereas elite climbers, who push the envelope both in the gym and on the rock, are likely to experience central fatigue. If you still feel physically “off” even after a few successive rest days, you may be experiencing central fatigue. It may take another two to ten days away from training and climbing to recuperate completely.

Three Recovery Periods

Recovery is not linear, but instead exponential (see figure 12.1). For example, recovery from an exhaustive crux sequence, climb, workout, or day of climbing will initially be rapid, with about 70 percent of complete recovery taking place in the first

Figure 12.1 Recovery Curve



Recovery to 90 percent of baseline levels occurs in about two-thirds of the time needed for complete recovery.

one-third of the recovery period (Bompa 1983). Recovery improves to 90 percent after two-thirds of the time needed for complete recovery.

Knowing that recovery is an exponential process is powerful information that can be applied to the three crucial recovery periods that I've defined as recharge, refuel, and rebuild.

RECHARGE (IMMEDIATE/SHORT-TERM RECOVERY)

This first recovery period takes place from ten seconds to sixty minutes following the completion of muscular action. The recharge period includes the ten-second shakeout you take in the midst of a crux sequence, as well as the mid-climb rest you milk for five, ten, or even twenty minutes, if possible.

The two metabolic processes at work during this recovery period are resynthesis of ATP-CP and the removal of H^+ , lactate, and other metabolic by-products from the working muscles. As stated earlier, ATP-CP resynthesis takes less than five minutes, and complete H^+ and lactate removal occurs in less than sixty minutes. As depicted in figure 12.1, however, recovery to 90 percent of baseline levels

occurs in about two-thirds of the time needed for complete recovery. Therefore, you can assume that the majority of ATP-CP resynthesis has occurred in just over three minutes and that blood pH has returned to near-normal levels (90 percent of H^+ removed) in about forty minutes. Strategies to further hasten recovery during the recharge period will be detailed later in this chapter.

REFUEL (MEDIUM-TERM RECOVERY)

The refuel recovery period occurs from sixty minutes to twenty-four hours following cessation of exercise. Therefore, this phase of recovery takes place during the two-hour break between climbs that you might take during the hottest part of the day (or while belaying your partner on their project) and, of course, during the overnight period following a training or climbing day.

With ATP-CP resynthesis and H^+ removal completed in the first sixty minutes, the refuel stage is defined by the replenishment of blood glucose and glycogen stores (in muscles and the liver), and some minor repair of tissue microtraumas. Since refueling is the hallmark of this period, consuming a large amount of the right type of carbohydrates is necessary to facilitate the process.

Numerous strategies for enhancing this replenishment process will be described later in the chapter, but—as mentioned earlier—you can assume that about 90 percent of this refueling has taken place in about sixteen hours (two-thirds of the total recovery period). Thus the typical twelve-hour break between consecutive days of climbing will result only in approximately 80 percent replenishment of glycogen stores (assuming you begin refueling immediately upon finishing up the first day on the rocks).

Rebuild (Long-Term Recovery)

Muscle growth and neuromuscular adaptation typically take place from one to four days following strenuous exercise. The degree of delayed-onset

muscle soreness experienced is proportionate to the amount of microscopic damage inflicted on muscle fibers during exercise. Minor DOMS may subside in forty-eight hours, while severe soreness signals a greater degree of damage that may take four or more days to rebuild.

You can now see that a single rest day is enough to allow the muscles to recharge and refuel, yet complete supercompensation—that is, rebuilding the muscle to a stronger level than before the exercise stimulus—requires additional rest for the neuromuscular system to recuperate. Therefore, while you may be able to perform at a reasonably high level after just one day of rest, truncating the rebuilding process negates the supercompensation period (see chapter 5) and will inhibit gains in strength. In the long term, chronic under-resting can result in decreased performance, injury, and even risk of illness.

Whether you are training indoors or climbing a personal-best route outside, the ability to accelerate recovery is tantamount to elevating your absolute level of performance. In the gym faster recovery between exercises or climbs translates to higher-intensity stimuli or faster learning, respectively. On the rock hastened recovery at a marginal rest may make the difference between a brilliant on-sight or hardest-ever flash and dangling on the rope in frustration.

On the pages that follow, you will learn fourteen powerful techniques for accelerating recovery and thus improving the quality of your training and climbing. But knowing is not enough—you must apply these strategies with the same dedication and resolve as when pursuing your endeavors in the gym or at the crag.

Accelerating Short-Term or Intraclimb Recovery

Your capacity to perform difficult moves or exercises repeatedly, with only short rest breaks, is directly

proportionate to your recovery ability in the short term. Described above as the recharge period, only certain recovery mechanisms come into play during the first ten seconds to sixty minutes following strenuous activity. As detailed in chapter 5, short-term recovery is driven by the aerobic energy system—therefore, your potential to recover quickly (in minutes rather than hours) is a direct function of your generalized aerobic fitness level. Anyway, your goal during this mid-climb or between-climb recovery period is to act in ways that will expedite the recharge process. This is done by minimizing the magnitude of fatigue (in the first place), engaging in steady, deep breathing, enhancing forearm recovery with the G-Tox (described below) and active rest, and engaging in pre-exercise hydration.

Limit Fatigue by Climbing More Efficiently

Let's start off with the most simple, yet powerful, method to enhance short-term recovery—limit the magnitude of fatigue, as much as possible, through economy of movement and optimal climbing technique. Obviously, you will use less ATP—CP and glycogen, as well as produce fewer metabolic by-products if you can lower the intensity of muscular contraction and the total time under load. In this way you immediately reduce the magnitude of the fatigue you must recover from, and you will return to baseline strength more quickly.

It's in this area that the average climber can realize a windfall of unknown capability. The fact is, most climbers move too slowly, possess less-than-ideal technique, and stop to place gear or think when they should be pushing on to the next rest. While lack of experience and technical ability are the real limiting factors here, slow climbing and hesitation will lead you to believe that a lack of strength is the primary problem.

If any of this sounds familiar, then some dedicated technique-training practice will go a long way toward elevating your game (see chapter 4). By learning to move swiftly and accurately through

hard moves, and by relaxing your grip and lowering tension in the antagonist muscles, you will use less glycogen on difficult sequences and reduce muscular acidosis and hypoxia. In this way you lower the magnitude of the fatigue you must recover from at a mid-climb rest position, and you will return to baseline strength more quickly between attempts or sends.

Enhance Forearm Recovery with the G-Tox

The dangling-arm shakeout is the technique traditionally used to aid recovery in the commonly fatigued forearm muscles. A few seconds or, hopefully, a few minutes of shaking out provides some recovery, but often not enough. The effects of a full-on pump can take frustratingly long to subside, and when hanging out at a marginal rest, it's possible to expend as much energy hanging on with one arm as is being recouped in the other. Such a zero-net gain in recovery does nothing to enhance

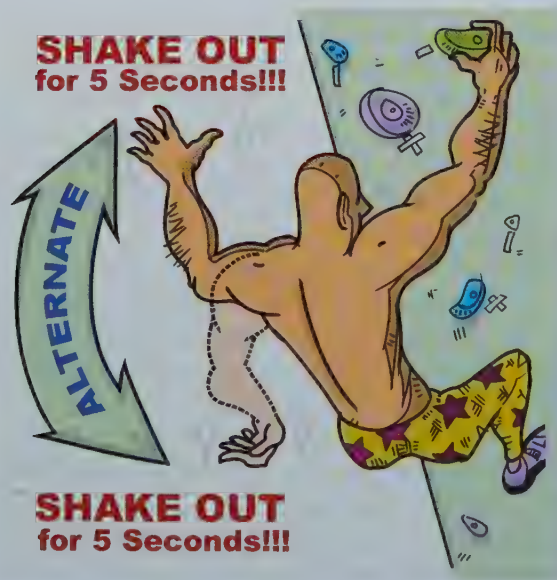
performance—in such a situation you would likely have fared better by blowing off the so-called rest and climbing onward.

Luckily there is the G-Tox, a shakeout technique that I developed to accelerate recovery of finger strength while hanging out at a mid-climb rest. For more than twenty years, I have been promoting the benefits of alternating the position of your resting arm between the normal dangling position and an above-your-head raised-hand position. This simple practice provides a noticeable increase in recovery rate—an effect pronounced enough to be confirmed in a 2003 university research project by Luke Roberts. I named this recovery technique the G-Tox, because it uses gravity (as an ally, for once) to help detoxify the fatigued muscle and speed recovery.

The discomfort and pump that develop in your forearms while climbing is largely the result of increasing intracellular acidosis and hypoxia in the working muscles. Contractions of as little as 20 percent of maximum intensity begin to hamper capillary blood flow, and at 50 percent of maximum contraction blood flow is largely occluded. When the contraction ends, blood flow resumes (reperfusion) at an accelerated rate to re-oxygenate the tissue and remove metabolic by-products. This reactive hyperemic response is what you recognize as “the pump.”

The pump effect is then amplified by the dangling-arm position (the traditional shakeout technique), because the rate of blood flow into the muscles may slightly exceed the rate of venous return (“old blood” exiting the muscle). This traffic jam of sorts perpetuates the pump and slows recovery, yet many climbers continue to dangle their arms and complain about how sickening a pump they have.

The G-Tox technique puts gravity to work by aiding venous return of blood toward the heart. By helping get blood out of the arm more quickly, this practice enhances the removal of H⁺ and other by-products and therefore returns you to a homeostatic



Research has shown the G-Tox technique to be effective in speeding forearm recovery at marginal rest positions.

ILLUSTRATION BY MIKE TEA

baseline more quickly. The effects of this technique are unmistakable—you will literally see the pump drain from the elevated arm due to the interesting fact that arterial flow into the arm is less affected by gravity than is venous return flow.

So why not just use the raised-arm position for the full duration of the rest instead of using the alternating technique as described above? Since the raised-arm position requires some muscular contraction in the upper arm, shoulder, and chest, these muscles would fatigue and possibly hamper climbing performance if you held the raised-arm position for a long time. Consequently, the best protocol for recovery is alternating between the two arm positions—raised-arm and dangling-arm—every five to ten seconds. Do so and you will definitely feel the difference the G-Tox makes!

Engage in Active Rest

Along with the G-Tox, active rest is another under-used yet highly effective strategy for accelerating recovery. While the G-Tox shines in its effectiveness to enhance recovery at a mid-climb shakeout, use of active rest between climbs is an equally effective strategy for increasing the rate of H⁺ and lactate removal from the working muscles and bloodstream.

Several recent studies, including one excellent study on climbers (Watts 2000), have shown that active rest significantly reduces blood lactate compared with the more common practice of passive rest. In the Watts study, fifteen advanced climbers attempted to redpoint a twenty-meter, 5.12b gym route, with eight of them engaging in active rest (recumbent cycling) and the others assigned to passive rest immediately following completion of the route. Periodic measurements of blood lactate revealed that the active-rest group returned to preclimb levels within twenty minutes, while the passive group took thirty minutes to return to baseline levels. Therefore, low-intensity active rest accelerated the clearing of lactate from the blood by almost 35 percent.

Applying this research finding at the crag is simple. Upon completing a pumpy route or redpoint attempt, instead of sitting down and resting passively (or worse yet, having a smoke), grab your water bottle and go for a casual twenty-minute hike. This will help accelerate the removal of H⁺ and lactate, as well as provide a mental break from the action. Both these factors will enhance your performance on the next route!

Another study compared recovery after maximum exercise in four groups: passive rest, active rest, massage, and combined massage and active rest (Monedero 2000). After fifteen minutes of rest, blood lactate removal was greatest in the group performing combined active rest and massage. Therefore, you may be able to further improve the Watts strategy of active rest by performing some self-massage on your most fatigued muscles (usually the forearms, upper arms, and shoulders).

A more recent study has shown that shorter periods of active recovery provide similarly positive effects on lowering blood lactate concentrations compared with equal periods of passive rest. The study tested ten climbers engaging in five, two-minute climbing trials, followed by two minutes of either active or passive recovery. The active-recovery group started the next trial with lower arterial lactate concentration than the passive-recovery group, and they recorded lower perceived exertion scores at the end of each climb (Draper 2006). The bottom line on this study is that in bouldering, it's better to walk around between ascents or attempts and sustain a higher heart rate to speed recovery.

Prehydrate Within Two Hours of Exercise

Muscles are more than 70 percent water, and it plays a vital role in cellular function and the transport of nutrients and metabolic waste. If you are dehydrated, it will hurt your performance and slow recovery. Therefore, it's prudent to prehydrate by consuming up to a quart of water in the hours preceding a workout or climbing. Continue sipping

Tips to Accelerate Short-Term Recovery

1. Strive to limit fatigue in the first place by climbing with maximum economy, optimal technique, and brilliant strategy!
2. Accelerate recovery at mid-climb shakeouts by using the G-Tox recovery technique.
3. Engage in active recovery between climbs by walking around and performing light stretching and massage. Research has shown that this accelerates recovery by up to 35 percent compared with just sitting around between ascents or attempts.
4. Do not smoke! Smoking decreases circulation and slows recovery. Worse yet, smoking inhibits the production of collagen and thus slows healing of tendons and ligaments.
5. Camel up by drinking a quart of water in the hours preceding a workout or day of climbing. Continue drinking water as you climb at a rate of eight ounces every hour or two.
6. Add moderate amounts of generalized aerobic exercise to your training-for-climbing program. A stronger cardiovascular system will provide faster recovery both at mid-climb rests and between boulder problems and routes.

water throughout the duration of activity at a minimum rate of eight ounces every hour or two (twice this if it's hot).

Accelerating Medium-Term or Intraday Recovery

Intraday recovery is the medium-term recuperation that occurs throughout the day and up to twenty-four hours following exercise. What you do (or don't do) during this recovery period plays a direct role in how much energy you will have

during the latter part of a long day of climbing; it's also the primary factor in how much recovery you can acquire in a single night of rest. Of course, this is of big-time importance if you are in the midst of a long, all-day route or when you plan to climb two days in a row.

Earlier I referred to this medium-term recovery phase as the refuel period, since restoring a normal blood glucose level and replenishing glycogen is the basis for most recovery gained from sixty minutes to twenty-four hours following exercise. Consequently, consuming the right carbohydrates at the right time is the single most vital action to accelerate recovery. Still, stretching, massage, foam rolling, and the use of relaxation exercises will also increase your rate of recuperation. Let's delve deeper into each of these areas.

Refuel Early and Often

The single biggest error in recovery strategy by most climbers is delayed consumption of calories during and after a day of climbing. The natural tendency is to become so engaged in the activity of climbing that you forget to eat and drink. This is compounded by the fact that strenuous exercise naturally suppresses hunger.

REFUELING WHILE YOU CLIMB

Earlier it was explained that consuming calories throughout the day helps maintain blood glucose and thus helps slow the use of your limited supply of glycogen. Toward this end, you should take in your first dose of calories between one to two hours after beginning your climb. If you are cragging, this might mean consuming a piece of fruit, a Protein Plus PowerBar, or eight ounces of sports drink after completing the first strenuous climb of the day. Continue eating a small serving of food every two hours throughout the day. In the case of all-day climbing, this means a total of four snacks—for example, two pieces of fruit and two energy bars.



Quality efforts—and project sends!—come when you take a proactive role in the recovery process. Make each attempt count, by beginning with fresh muscles and a focused mind ready to try hard!

BRUCE WILSON

This may seem like an awful lot of food, and it is if you are only climbing for half a day or going bouldering (halve these amounts in those situations). To keep climbing hard throughout the day and to speed recovery for a second day of climbing, however, you should consume a minimum of 600 to 800 calories during the course of the day.

Selecting the right kinds of food at the right time is a matter of using the glycemic index (GI). As introduced in chapter 11, high-GI foods elicit a rapid rise (then drop) in blood sugar, while medium- and low-GI foods release fuel into the bloodstream more slowly (see table 11.2). In a stop-and-go sport like climbing, steady blood sugar is vital for maintaining steady concentration and steady energy. Therefore, consume only low- and medium-GI foods while you are still engaged in

physical activity. Upon completing your day on the rocks or reaching the end of your workout, however, it's best to consume higher-GI foods and beverages. This latter distinction is powerful—there is a growing body of research indicating that what you eat in the first thirty minutes after exercise is the single largest determining factor in how fast you recover.

KICK-STARTING GLYCOGEN REPLENISHMENT AFTER CLIMBING OR TRAINING

As incredible as it may seem, recent research has shown that waiting two hours after exercise to consume carbohydrates will reduce your glycogen replenishment by 50 percent compared with eating immediately upon cessation of the activity (Burke 1999). Therefore, when planning to climb a second

day, you significantly handicap tomorrow's performance by delaying refueling. Similarly, delayed refueling after training slows the recovery and rebuilding processes and, possibly, delays complete recovery by as much as a full day.

Let's take a closer look at the best refueling strategies in the hours following climbing or a vigorous workout.

First thirty minutes after climbing: Ingestion of high-GI foods immediately after exercise substantially increases the rate of muscle glycogen replacement (Richter 1984). More recent studies have shown that glycogen resynthesis may take place another 40 percent faster if protein and carbohydrate are consumed together, due to a greater insulin response (Niles 1997). Consequently, the best protocol for accelerating glycogen replenishment appears to be a 4:1 ratio of carbohydrate to protein consumption (Burke 1999).

Since solid foods enter the bloodstream more slowly than liquids, it's best to drink this carbohydrate-protein blend post-exercise. For example, a 160-pound climber would want to consume approximately one hundred grams of carbohydrate and twenty-five grams of protein. Drinking a pint of Gatorade, juice, or another glucose- or high-fructose-corn-syrup-based sports drink would provide about fifty grams of high-GI carbohydrate, while consuming a high-protein energy bar or a whey protein shake would provide around twenty grams of protein. Another highly practical option is to drink a 4:1 (carbohydrate-to-protein ratio) sports drink such as Accelerade or a pint of chocolate skim milk. Yes, chocolate skim milk is a most excellent post-exercise recovery drink, considering that it contains over fifty grams of carbohydrate and seventeen grams of high-quality protein.

Two hours after climbing: Assuming you consumed the initial feeding of carbohydrates and protein within the thirty-minute time frame, you can wait until about two hours post-activity to eat a complete meal. Ideally, the meal will include foods

providing a macronutrient ratio of about 65:15:20 (calories from carbohydrate, protein, and fat, respectively), as explained in chapter 11. Such a meal might include a moderate serving of pasta, a piece of lean meat, and a large salad or some vegetables. Whereas high-GI foods are best eaten immediately after exercise, medium- and low-GI foods are more advantageous in the two to twenty-four hours after exercise. They will provide a slower, longer-lasting trickle of glucose into the bloodstream that will support steady glycogen resynthesis.

Before going to sleep: A small meal of carbohydrate and protein within thirty minutes of going to sleep will further support glycogen resynthesis and tissue remodeling overnight. Skim milk may be the perfect before-bedtime food—it possesses low-GI carbohydrate, high-quality protein, and the amino acid tryptophan, a precursor to serotonin, which slows down brain activity. Drink a tall glass of skim milk (consider mixing in some whey protein powder) or have some skim milk on a small bowl of whole-grain cereal before bedtime. It'll do your body good!

Stretch and Massage the Hardest-Working Muscles

Earlier, you learned about an impressive research study showing that combining active rest and massage accelerates recovery by enhancing the removal of metabolic by-products from the working muscles. Deep-fiber-spreading sports massage is also an effective practice to enhance medium- and long-term recovery.

HOW IT WORKS

Traditional massage has long been used to increase blood flow and oxygen transport in the muscles. The benefits of this superficial "rubbing" are brief, however, and have little residual effect on performance.

Sports massage utilizes a deep-fiber-spreading technique that produces hyperemia (vasodilation)

through the full depth of the muscle. Furthermore, the state of hyperemia lasts long after the procedure has ended, so the enhanced blood flow can accelerate recovery and aid healing after strenuous exercise.

Sports massage also helps reduce the number of small and often unfelt spasms that regularly occur in the muscles. These spasms may go unchallenged by conventional stretching and warm-up exercises and, left unchecked, can rob you of coordination and induce mechanical resistance and premature fatigue.

HOW TO DO IT

There are several strokes you may want to learn, but the most effective is called cross-fiber friction. This stroke is best executed with a braced finger (see photo). The motion is a simple push in followed by a short push back and forth across the muscle fiber. Keep the stroke short and rhythmic; only gradually increase the pressure to penetrate deeper into the muscle.



Cross-fiber friction (with braced finger) self-massage is an excellent way to increase blood flow and warm up the forearm muscles prior to training. Armaid's Rubbit (see page 122) is a wonderful invention, as it provides additional leverage for attacking trigger points in the forearms.

Although sports massage can be used on all muscles, focus your efforts on the upper body and, in particular, the finger flexors and extensors, the pronator teres and brachioradialis (upper forearm muscles), the biceps and triceps (muscles of the upper arm), and the large muscles of the shoulders and back. Incorporate five to ten minutes of massage into your regular warm-up routine. This, along with the warm-up activities and stretches detailed in chapter 6, will better prepare you for an excellent workout or day on the rocks.

SPORTS MASSAGE TO ACCELERATE RECOVERY

Your body has inherent mechanical weaknesses where sport-specific movements can trigger stress overload. In climbing, the muscles most overloaded are the forearms, upper arms, and back. These muscles are the first to tire, and they are typically the slowest to recover. Fortunately, you can modulate fatigue and hasten recovery through application of sports massage to the specific stress points—often called *trigger points*—inherent to climbing movements.

Use of sports massage and trigger point therapy will help relieve these stress points and speed recovery. You can best address these trigger points with what is called the direct-pressure stroke. Apply firm, constant, straight-in pressure with a braced finger, an Armaid device, a wooden Bodo or shepherd's crook, or a friend's elbow, and hold for fifteen to thirty seconds. Direct pressure is especially useful when applied to the trigger points near the base of the muscles. This will help relieve any known or unknown spasms, increase local circulation, and aid healing. (Be sure never to apply sports massage tactics to tendons, joints, or injured tissues, however.) Conclude your massage with some foam rolling and mild stretching.

Use Relaxation Techniques

Chapter 3 described an excellent relaxation technique called the Progressive Relaxation Sequence.

Tips to Accelerate Medium-Term Recovery

1. Refuel early and often. Consume a piece of fruit, a balanced-style energy bar, or eight ounces of sports drink every hour or two throughout the day of climbing. Favor low- to medium-GI foods and drinks.
2. At the end of a workout or day of climbing, kick-start recovery by consuming a high-GI sports drink or a 4:1 ratio carbohydrate-to-protein sports drink within the first thirty minutes of cessation of strenuous activity. Chocolate skim milk is an excellent choice!
3. Within two hours of concluding your workout or climbing, consume a meal comprising approximately a 65:15:20 percent breakdown of carbohydrate, protein, and fat, respectively.
4. Avoid consumption of alcoholic beverages, since alcohol has been shown to slow super-compensation, reduce beneficial growth hormone response, and possibly even slow protein synthesis.
5. Stretch, massage, and use a heating pad or whirlpool to increase blood flow to sore, tired muscles. Use cross-fiber friction massage, foam rolling, and direct pressure to relieve trigger point pain and spasms.
6. Engage in progressive relaxation (see chapter 3) to further relax muscles and quiet the mind during midday breaks from climbing and before going to sleep.

Though commonly used before going to sleep, progressive relaxation is also highly effective for relaxing the muscles and quieting the mind during a midday break from climbing. When resting between climbs or taking a break before returning to work on a project, find a quiet spot, lie down, and spend

ten to twenty minutes performing progressive relaxation. Upon completing this process, sit up for a few minutes and enjoy the day before proceeding to the next climb.

Make a midday relaxation break a regular part of your climbing ritual and you'll find yourself climbing better, with less fatigue, late into the day.

Accelerating Long-Term or Interday Recovery

The interday recovery period involves the long-term recuperation from a severe workout or a couple of hard days of climbing. Depending on the intensity and volume of the activity, full recovery could take anywhere from one to four days.

When you wake up in the morning with sore muscles (delayed-onset muscle soreness), it's a sign that you incurred microtraumas and that a recovery period of at least another twenty-four hours is needed. Of course, you have two choices in this situation. The first is to go climbing (or work out) for a second straight day, despite the soreness, realizing that your performance will be less than ideal and your risk of injury may be increased. Or you could take a day or two off and allow your neuromuscular system to recuperate to a level of capability higher than before the workout.

Certainly, there are times when you will select the first option of climbing a second day straight, but there should be an equal number of instances when you decide that "less will be more." Weekend climbing trips are the classic situation in which you'd want to climb two days in a row, regardless of sore muscles. Given proper nutrition, a good warm-up, and a prudent approach to pushing yourself on the second day, you can usually get away with climbing sore.

Choosing to take a day or two of rest, however, is clearly the intelligent decision when climbing indoors or during an off-season training cycle. Hopefully, you gleaned from chapter 5 that proper rest is as important as training stimulus in becoming

a stronger climber, and that under-resting is a primary cause of injury. Enthusiastic indoor and sport climbers are most commonly guilty of under-resting, but regardless of your climbing preference, it's important that you distinguish yourself from the mass of climbers who overtrain. If you find yourself drawn to overtraining with the crowd, remember that in order to outperform the masses, you must act in ways that they do not!

Eat Frequent, Small Meals

Instead of eating the typical three meals per day, you can accelerate recovery by consuming six smaller meals or snacks spaced evenly throughout the day. Avoid high-GI foods; they are less effective for recovery after the first two hours post-exercise. Instead select low- and medium-GI foods for all your meals, and drink at least two quarts of water throughout the day.

At least three of your meals should contain a significant portion of protein. For instance, breakfast could include a couple of eggs, skim milk, or whey protein; lunch might include some low-fat yogurt, skim milk, or a can of tuna; and for dinner it might be good to eat a piece of lean red meat, chicken, or fish. Each of these meals should also include some carbohydrate, and at all costs avoid fat-laden fried foods and any snack foods containing hydrogenated oils. Strive for a macronutrient profile of roughly 65:15:20 (carbohydrate:protein:fat) for each major meal.

While the other three feedings need be only a couple hundred calories, they are vital for maintaining steady blood glucose and continuing the recovery processes throughout the day. Low- and medium-GI foods are the best choice, with a piece of fruit or a balanced-type energy bar being ideal selections.

Take a Multivitamin

In today's world of highly processed foods, it's often difficult to consume enough of the vitamins and

minerals that athletes need by simply eating a well-rounded diet. Consider taking a daily multivitamin as a must; additional magnesium is beneficial in a number of ways and may even help your muscles relax and return to normal function.

Stretch, Foam Roll, and Massage Sore Muscles

Gentle stretching, foam rolling, and sports massage are widely accepted as effective means to lengthen the muscles and enhance recovery following strenuous exercise. Professional athletes have full-time trainers who help with post-exercise stretching and rest-day massage (must be nice!). Though I won't pay for you to add a masseur to your climbing-support staff, I do suggest that you stretch and foam roll for at least ten to twenty minutes per day. Follow the stretching and foam rolling procedures outlined in chapter 6 and, of course, the primer on sports massage provided earlier in this chapter.

Get Plenty of Sleep

Here's an important recovery technique that this full-time working, four-day-a-week-training, weekend-climbing, book-writing, forty-year veteran of the rocks wishes he could get more of! Although most neuromuscular regeneration occurs during sleep, I always fall back on the fact that nothing is produced or achieved during sleep. But I digress.

Seriously, sleep is vital for any climber serious about training and passionate about maximizing ability. The bare minimum amount of sleep per night is six to seven hours, though eight to ten hours is ideal following an extremely strenuous workout or exhaustive day of climbing. No doubt, it's a busy world—and sleep may seem like the only activity that's expendable. If you closely evaluate a typical day, however, you will likely be able to identify some low-value activities like surfing the Net, watching TV, and certain social events that can be reduced or eliminated to allow for more sleep. It

Long-Term Recovery Tips

1. Eat small, frequent meals—consume six smaller meals and snacks comprising low- to mid-GI foods and protein sources.
2. Drink eight ounces of water every one to two hours to rehydrate and flush toxins from your body.
3. Take a daily multivitamin; supplemental magnesium is also highly recommended.
4. Avoid excessive consumption of alcoholic beverages, since these provide little quality nutrition, enhance dehydration, and in high doses suppress growth hormone response and protein synthesis following exercise.
5. Engage in daily stretching, foam rolling, massage, and active-recovery activities such as hiking, biking, and easy running.
6. Apply a heating pad to sore muscles, twice daily for ten to fifteen minutes, to increase blood flow and accelerate recovery and healing. Note that you should never place a heating pad on a swollen or acutely injured body part.
7. Get at least six to seven hours of sleep per night, eight to ten hours (or more) after mega-long day climbs or multiday big-wall or alpine ascents.
8. Strive for a calm, relaxed, easygoing disposition every day—this will help lower cortisol levels and foster more rapid recovery.

can take great discipline to give up some of these activities—visualize your climbing goals!—but the long-term payoffs will dwarf the hollow pleasure of these low-value pastimes.

Engage in Light Activities

Earlier you learned of a couple of great research studies that showed the value of active rest in accelerating recovery from strenuous exercise. In the

context of long-term recovery, active rest is also beneficial because it enhances circulation of nutrients to the damaged muscles and produces a general loosening effect on stiff muscles.

The best active-rest activities for climbers are hiking, jogging, light mountain biking, and even some limited less-than-vertical or big-hold climbing (ARC training protocol). Still, it's crucial that each of these activities be performed at a low enough intensity that you don't get heavily winded or pumped and break only a light sweat. Limit yourself to thirty to sixty minutes of active-recovery exercise, and remain disciplined in not letting the activity escalate into anything more than active rest.

Possess a Positive, Calm Personality Each and Every Day

This last recovery tip is subtle yet very powerful. Possessing a positive, relaxed, and easygoing attitude not only puts you in a better performance state, but it also has been proven to increase recovery and maybe even encourage muscular growth.

Strenuous exercise and stressful situations cause a drugstore's worth of chemicals and hormones to be released into the bloodstream. Some of these hormones have long-term positive effects, such as growth hormone, which is anabolic. The fight-or-flight hormones like epinephrine and cortisol, however, can have a long-term negative effect when released chronically. In particular, cortisol has been shown to be catabolic, meaning that it results in breakdown of muscle.

In light of the above factors, elite athletes have long been interested in enhancing the release of growth hormone and preventing high levels of cortisol. This is the very reason some athletes take anabolic steroids.

Fortunately, you can modulate levels of growth hormone and cortisol with proper training, quality nutrition, and adequate rest, as well as through adjustments in your lifestyle. For instance, individuals with Type A, aggressive behavior naturally

exhibit higher levels of cortisol (Williams 1982) and reduced levels of growth hormone. It's also been shown, however, that behavior modification and reduction of the stressors in life can reverse this effect and provide more beneficial training (Dinan 1994). Therefore, possessing a relaxed approach to climbing and a humorous attitude about life in general will play an underlying but beneficial role by enhancing the quality of your training adaptations, accelerating recovery, and boosting climbing performance.

It's important to recognize that training and recovery are opposite sides of the same coin. You must place equal importance on doing both optimally and to the best of your ability. Clearly, it requires a shift in perspective to actually plan and engage in the process of recovery in the same way you plan and engage in the process of training. But in doing so, you will distinguish yourself from the masses by producing uncommonly good results, and by avoiding downtime due to injury and illness.



*Inge Perkins on Bobcat Logic
(12b/c), Wild Iris, Wyoming.*

SAVANNAH CUMMINS